

RETAIL SALES & PROFABILITY ANALYSIS DASHBOARD & REPORT

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Tools used: RStudio, Tableau.

Project Overview:

In this project, I use R to analyze and clean data from an online sports store. The goal is to answer key business questions and find useful insights about revenue, profit, customer ratings, and sales by state.

Additionally, I created an interactive Tableau dashboard to better understand the data. You can explore it here: <https://public.tableau.com/app/profile/sebastian.ortuno.barrero/viz/SportsSalesCustomerAnalysis/RatingsDashboard?publish=y>

This project tries to solve an important business problem: understanding what parts of the business are performing well and what parts need improvement. By looking at sales, profit, customer ratings, and location data, I try to find which sports bring the most value, which states have better results, and where the company may have opportunities to grow.

This project also tries to solve the problem of limited visibility in business performance over time. By studying monthly revenue and profit trends, I try to find patterns during the year, compare strong and weak months, and understand customer behavior better. These results can help the business make better decisions about marketing, sales, and future planning.

Data Cleaning and Business Requirements:

- 1) Load libraries
 - 2) Data Exploratory
 - 3) Convert 'date' column (in text format) to a proper DATE type and store in 'Date_New'.
 - 4) KPIs: total revenue, profit, number of orders, profit margin.
 - 5) KPIs by sport: revenue, profit, orders, profit margin.
 - 6) Customer ratings: number, the percentage of ratings the company got from all the orders, average rating.
 - 7) Ratings distribution: number of orders by rating, revenue by rating, profit by rating, and profit margin by rating.
 - 8) Revenue, profit, and profit margin by State.
 - 9) Monthly profit trends and comparison with previous month.
 - 10) Monthly revenue trends and comparison with previous month.
 - 11) Executive Summary
-

1) Load Libraries:

Before we start with the business requirements, I load the libraries needed for this project.

```
library(openxlsx)
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.1
```

```
## Warning: package 'ggplot2' was built under R version 4.5.1
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2     4.0.0      v tibble     3.2.1
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.0.4
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
library(lubridate)
library(ggplot2)
library(knitr)
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```
library(scales)
```

2) Data Exploratory

After loading the data, we begin our analysis by using the `head()` and `glimpse()` functions to explore the structure and content of the dataset.

There are two tables: **orders** and **customers**.

- The **orders** table contains 8 columns and 2,847 rows. The columns are: `order_id`, `customer_id`, `sport`, `revenue`, `profit`, `shipping_cost`, `rating`, and `date`.
- The **customers** table contains 4 columns and 2,847 rows. The columns are: `customer_id`, `full_name`, `email`, and `state`.

```
# Load the orders dataset from the local project directory
orders <- read.csv("C:/Users/sebas/OneDrive/Documents/NEC MASTERS/Projects Portfolio/Projects Portfolio.

# Load the customer dataset containing customer information
customer <- read.csv("C:/Users/sebas/OneDrive/Documents/NEC MASTERS/Projects Portfolio/Projects Portfol.

# Preview the first rows of the orders dataset
head(orders)
```

```
##      date order_id customer_id      sport revenue profit shipping_cost rating
## 1 1/1/2022   10001     102278   Baseball  183.60  97.29             0     NA
## 2 1/1/2022   10002     102279 Basketball  185.76 103.40             0     NA
## 3 1/1/2022   10003     102280 Basketball  128.16  66.27             0     NA
## 4 1/1/2022   10004     102281     Hockey   45.62  15.46             7     NA
## 5 1/1/2022   10005     102282   Football  106.30  21.75             0     NA
## 6 1/1/2022   10006     102283   Football   58.11  12.08             0      3
```

```
# Preview the first rows of the customer dataset
head(customer)
```

```
##   customer_id    full_name      email      State
## 1    102278    Alica Reary areary0@sciencedaily.com    Florida
## 2    102279 Delmor Rubin   drubin1@yahoo.co.jp    Indiana
## 3    102280   Joanie Hoyt   jhoyt2@bloglovin.com Pennsylvania
## 4    102281 Madelena Boat   mboat3@surveymonkey.com    Nevada
## 5    102282 Sayers Patkin   spatkin4@sogou.com    New York
## 6    102283 Merwyn Stout    mstout5@sfgate.com    Michigan
```

```
# Inspect the structure and variables of the orders dataset
glimpse(orders)
```

```
## Rows: 2,847
## Columns: 8
## $ date      <chr> "1/1/2022", "1/1/2022", "1/1/2022", "1/1/2022", "1/1/202~
## $ order_id  <int> 10001, 10002, 10003, 10004, 10005, 10006, 10007, 10008, ~
## $ customer_id <int> 102278, 102279, 102280, 102281, 102282, 102283, 102284, ~
## $ sport     <chr> "Baseball", "Basketball", "Basketball", "Hockey", "Footb~
## $ revenue   <dbl> 183.60, 185.76, 128.16, 45.62, 106.30, 58.11, 214.04, 37~
## $ profit    <dbl> 97.29, 103.40, 66.27, 15.46, 21.75, 12.08, 70.41, 2.55, ~
## $ shipping_cost <int> 0, 0, 0, 7, 0, 0, 0, 7, 0, 0, 7, 0, 0, 0, 0, 7, 7, ~
## $ rating    <int> NA, NA, NA, NA, NA, 3, 2, NA, 2, NA, 5, 2, 2, NA, NA, 2, ~
```

```
# Inspect the structure and variables of the customer dataset
glimpse(customer)
```

```
## Rows: 2,847
## Columns: 4
## $ customer_id <int> 102278, 102279, 102280, 102281, 102282, 102283, 102284, 10~
## $ full_name   <chr> "Alica Reary", "Delmor Rubin", "Joanie Hoyt", "Madelena Bo~
## $ email       <chr> "areary0@sciencedaily.com", "drubin1@yahoo.co.jp", "jhoyt2~
## $ State       <chr> "Florida", "Indiana", "Pennsylvania", "Nevada", "New York"~
```

3) Convert 'date' column (in text format) to a proper DATE type and store in 'New_date'.

```
# Convert the 'date' column to a Date type and create a new column 'New_date'
orders<-orders%>%mutate(New_date=as_date(date, format="%m/%d/%Y"))
# Select and reorder columns, replacing 'date' with the new 'New_date' column
orders<-orders%>%select("order_id","customer_id","sport","revenue","profit","shipping_cost",
                        "rating","New_date")
```

4) KPIs: total revenue, profit, number of orders, profit margin.

```
# Calculate main business metrics from the orders dataset
KPI <- orders %>%      # Use the orders table
  summarize(          # Create summary metrics
    Total_Revenue = sum(revenue),      # Total revenue from all orders
    Total_Profit = sum(profit),         # Total profit from all orders
    N_orders = n_distinct(order_id),    # Count the number of unique orders
    Profit_Margin = round((Total_Profit / Total_Revenue) * 100, 2) # Profit margin (%)
```

```
)

# Display the results in a formatted table
kable(KPI)
```

Total_Revenue	Total_Profit	N_orders	Profit_Margin
459418.4	284821.9	2847	62

- **Total Revenue:** \$459,418.40
The store generated nearly half a million in total sales — strong revenue.
- **Total Profit:** \$284,821.90
Profit makes up a significant portion of revenue, indicating healthy operations.
- **Number of Orders:** 2,847
On average, each order generates about \$161.35 in revenue
($459,418.4 \div 2,847$)
- **Profit Margin:** 62%
Very high margin — over half of every dollar earned is profit, which is excellent for retail.

5) KPIs by sport: revenue, profit, orders, profit margin.

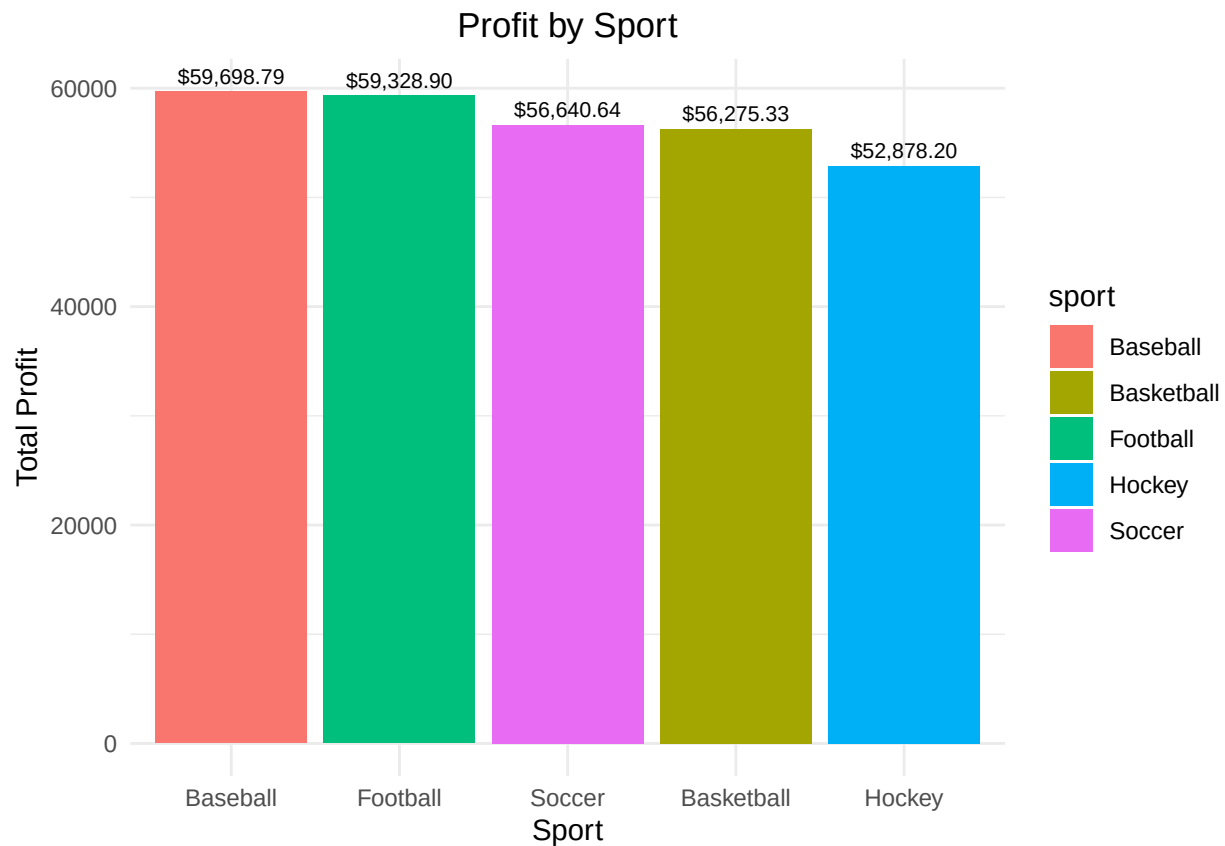
```
# Create KPI metrics by sport category
KPI_sports <- orders %>%           # Start with the orders dataset
  group_by(sport) %>%             # Group data by sport
  summarise(
    Total_Revenue = sum(revenue),  # Total revenue for each sport
    Total_Profit = sum(profit),     # Total profit for each sport
    Total_Orders = n_distinct(order_id), # Count unique orders
    Profit_Margin = round((Total_Profit / Total_Revenue) * 100, 2) # Profit margin (%)
  ) %>%
  arrange(desc(Total_Profit), desc(Total_Orders)) # Sort by profit and number of orders

# Show the results as a table
kable(KPI_sports)
```

sport	Total_Revenue	Total_Profit	Total_Orders	Profit_Margin
Baseball	95363.99	59698.79	565	62.60
Football	94768.32	59328.90	572	62.60
Soccer	90157.93	56640.64	561	62.82
Basketball	92116.23	56275.33	577	61.09
Hockey	87011.90	52878.20	572	60.77

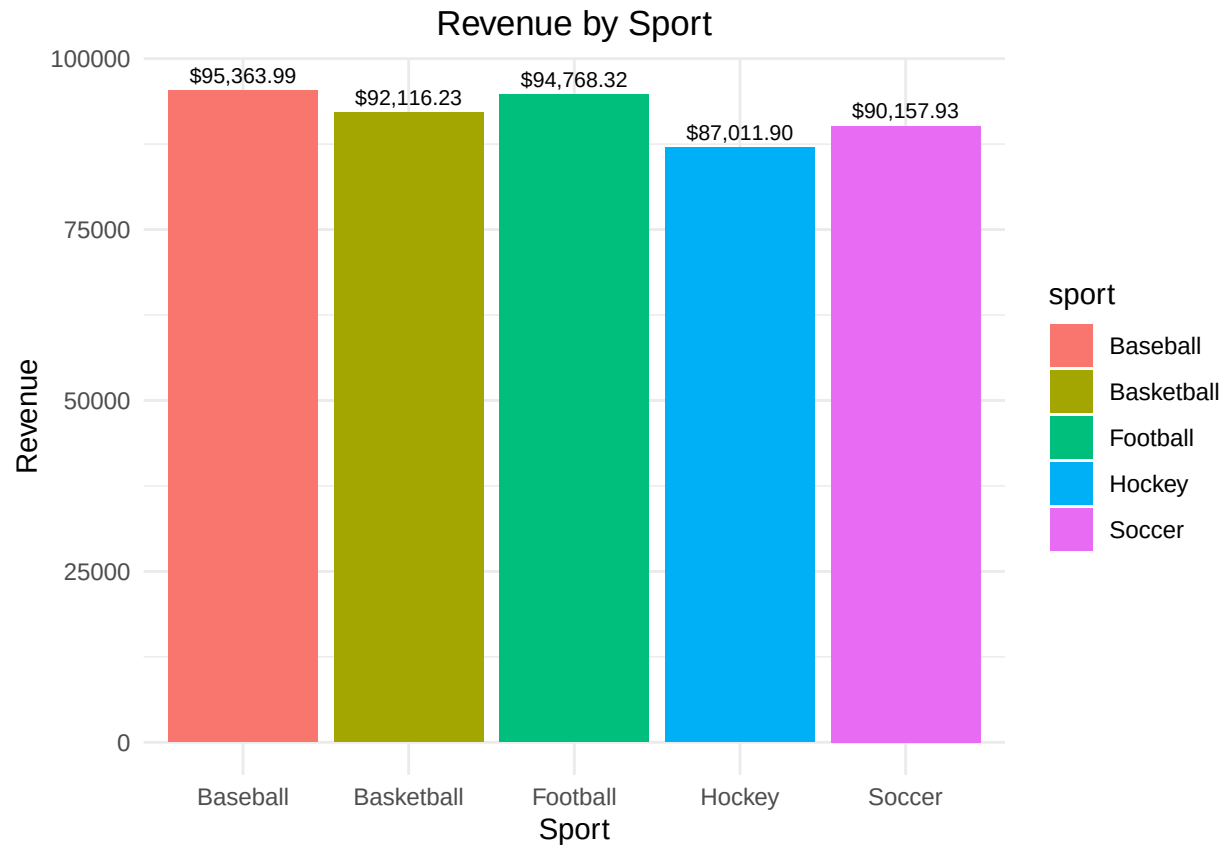
5.1 Profit by sports.

```
# Create a bar chart showing profit by sport
ggplot(KPI_sports, aes(x = reorder(sport, -Total_Profit), y = Total_Profit, fill = sport)) +
  geom_bar(stat = "identity") + # Create bars using total profit values
  geom_text(aes(label = dollar(Total_Profit)), # Add profit labels on top of bars
            vjust = -0.5, size = 2.7) +
  labs(title = "Profit by Sport", # Chart title
        x = "Sport", # Label for x-axis
        y = "Total Profit") + # Label for y-axis
  theme_minimal() + # Use a clean minimal theme
  theme(plot.title = element_text(hjust = 0.5)) # Center the title
```



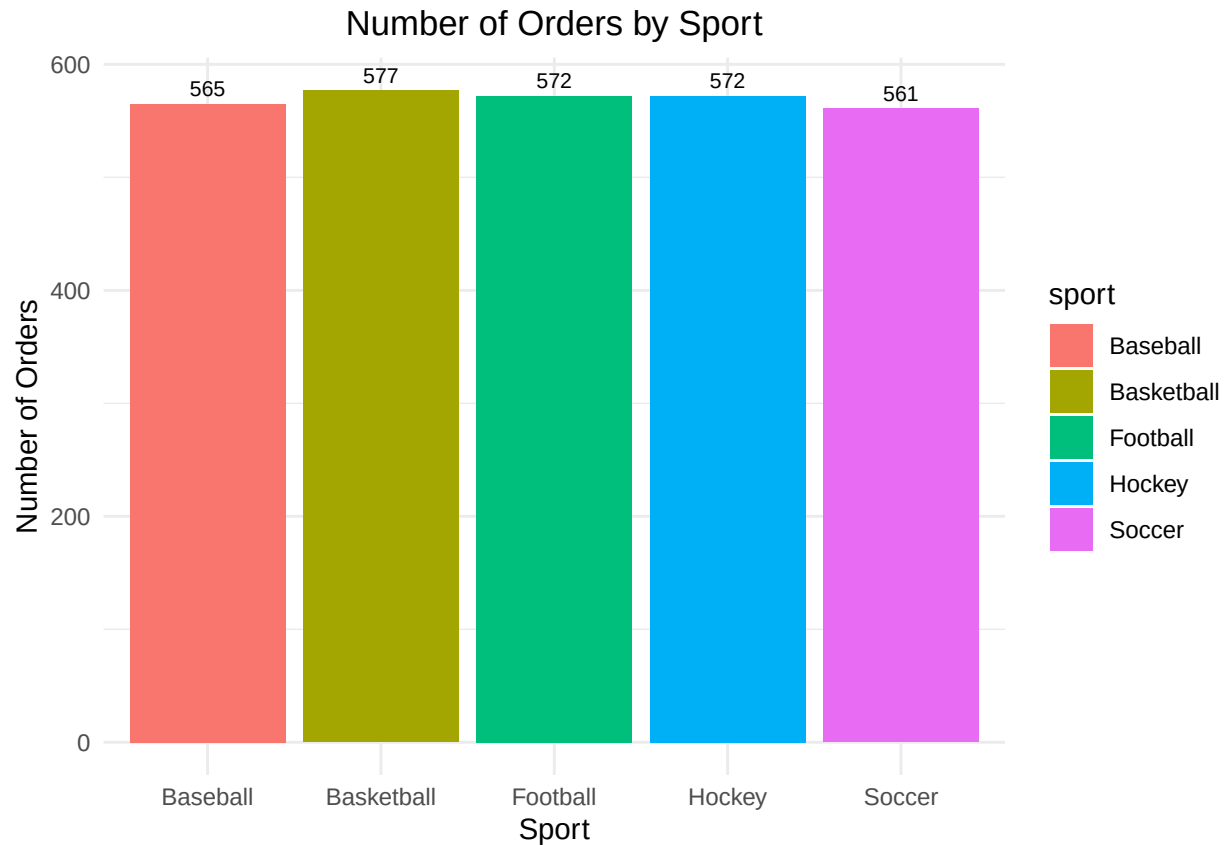
5.2 Revenue by Sport

```
# Create a bar chart showing total revenue by sport
ggplot(KPI_sports, aes(x = sport, y = Total_Revenue, fill = sport)) +
  geom_bar(stat = "identity") + # Draw bars using the revenue values
  geom_text(aes(label = dollar(Total_Revenue)), # Add revenue labels on top of bars
            vjust = -0.5, size = 2.8) +
  labs(title = "Revenue by Sport", # Chart title
        x = "Sport", # Label for x-axis
        y = "Revenue") + # Label for y-axis
  theme_minimal() + # Use a clean theme
  theme(plot.title = element_text(hjust = 0.5)) # Center the chart title
```



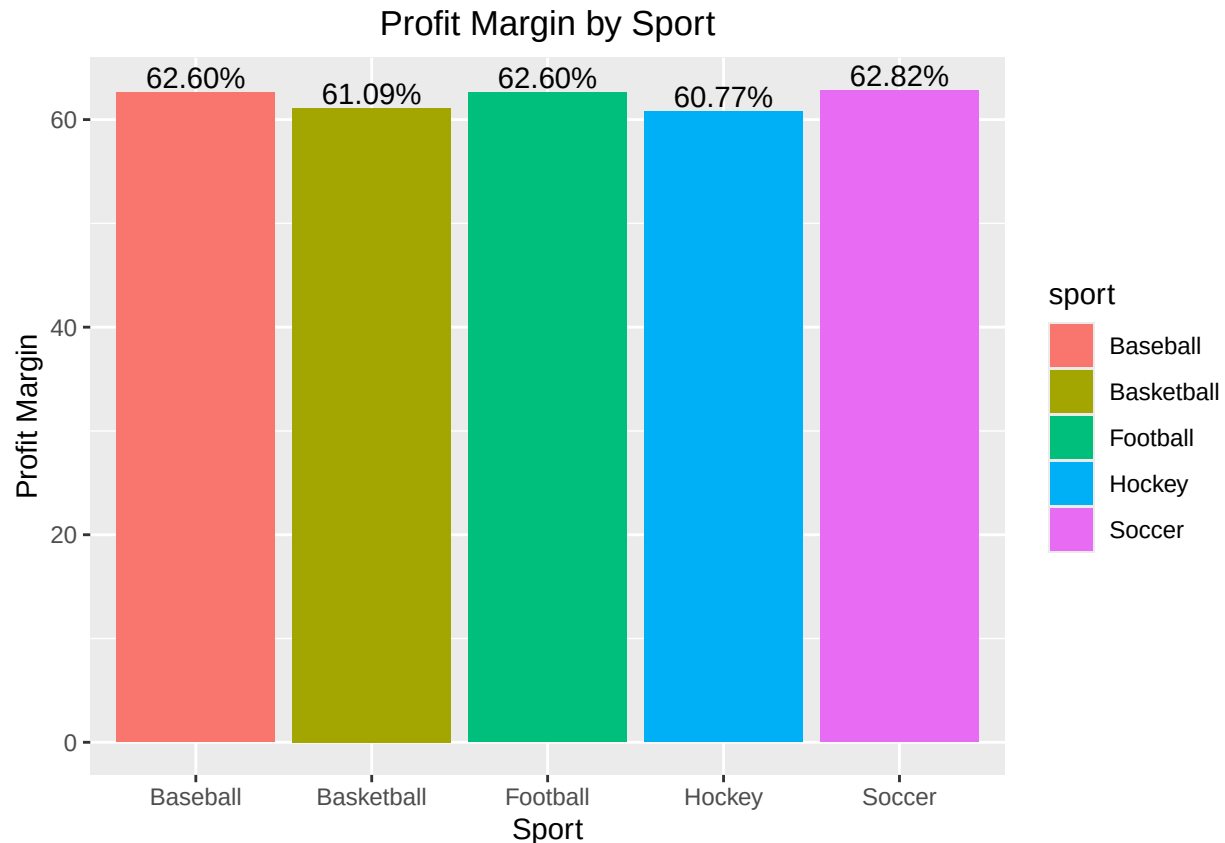
5.3 Number of orders by Sport.

```
# Create a bar chart showing the number of orders by sport
ggplot(KPI_sports, aes(x = sport, y = Total_Orders, fill = sport)) +
  geom_bar(stat = "identity") + # Draw bars using the number of orders
  geom_text(aes(label = Total_Orders), # Add order count labels above the bars
            vjust = -0.5, size = 2.8) +
  labs(
    title = "Number of Orders by Sport", # Chart title
    x = "Sport", # Label for x-axis
    y = "Number of Orders" # Label for y-axis
  ) +
  theme_minimal() + # Use a simple clean theme
  theme(plot.title = element_text(hjust = 0.5)) # Center the title
```



5.4 Profit Margin by Sport

```
# Create a bar chart showing profit margin by sport
ggplot(KPI_sports, aes(x = sport, y = Profit_Margin, fill = sport)) +
  geom_bar(stat = "identity") + # Draw bars using the profit margin values
  geom_text(aes(label = percent(Profit_Margin / 100)), # Add profit margin labels
            vjust = -0.2) +
  labs(
    title = "Profit Margin by Sport", # Chart title
    x = "Sport", # Label for x-axis
    y = "Profit Margin" # Label for y-axis
  ) +
  theme(plot.title = element_text(hjust = 0.5)) # Center the chart title
```

5.5 Comments:

- All sports have profit margins above 60%, indicating strong overall profitability.
- **Soccer** has the highest profit margin, while **Basketball** leads in order volume.
- **Football** and **Baseball** show an excellent balance between high revenue and strong margins.
- **Hockey**, although slightly lower in margin, still performs well and could benefit from cost optimization strategies.
- The number of orders is relatively consistent across all sports categories.
- **Recommendation:**
Consider reallocating marketing resources to **Hockey** and **Baseball** to boost volume, as they have strong performance potential. Additionally, analyze Basketball's pricing and cost structure to increase margins without sacrificing volume.

6) Customer ratings: number, percentage of ratings from all orders, average rating.

```
# Calculate customer rating metrics
Customer_Ratings <-
  orders %>% # Start with the orders dataset
  summarise(
    Average_Rating = round(mean(rating, na.rm = TRUE), 2), # Calculate average rating (ignore missing)
    Total_Rating = sum(!is.na(rating)), # Count how many ratings exist
```

```

Percentage_rating = round((Total_Rating / 2847) * 100, 2) # Percentage of orders with ratings
)
# Display the results in a table
kable(Customer_Ratings)

```

Average_Rating	Total_Rating	Percentage_rating
3.13	1193	41.9

- **6.1 Average Rating: 3.13**

The average customer rating across all orders is 3.13 out of 5, indicating a moderate level of customer satisfaction.

- **6.2 Total Ratings Submitted: 1,193**

Out of 2,847 total orders, 1,193 included a customer rating.

- **6.3 Percentage of Orders with Ratings: 41.9%**

Only 41.9% of customers submitted a rating, suggesting that the business may benefit from encouraging more feedback to better evaluate customer experience.

- **6.4 Recommendation:**

Implement post-purchase incentives such as small discounts, loyalty points, or thank-you emails to encourage more customers to leave ratings. Increasing feedback volume will provide better insights for improving products and service.

7) Ratings distribution: number of orders by rating, revenue by rating, profit by rating, and profit margin by rating.

```

# Analyze orders by rating category
rating_distribution <- orders %>% # Start with the orders dataset
  mutate(
    rating = as.character(rating), # Convert rating to text
    rating = replace_na(rating, "No rating"), # Replace missing ratings with "No rating"
    rating = factor(rating, levels = c("5", "4", "3", "2", "1", "No rating")) # Set rating order
  ) %>%
  group_by(rating) %>% # Group data by rating value
  summarise(
    orders_by_ratings = n(), # Count number of orders per rating
    revenue_by_rating = sum(revenue), # Calculate total revenue per rating
    profit_by_rating = sum(profit), # Calculate total profit per rating
    profit_margin = round((profit_by_rating / revenue_by_rating) * 100, 2) # Profit margin (%)
  ) %>%
  arrange(desc(rating)) # Sort ratings from highest to lowest
# Display the results as a table
kable(rating_distribution)

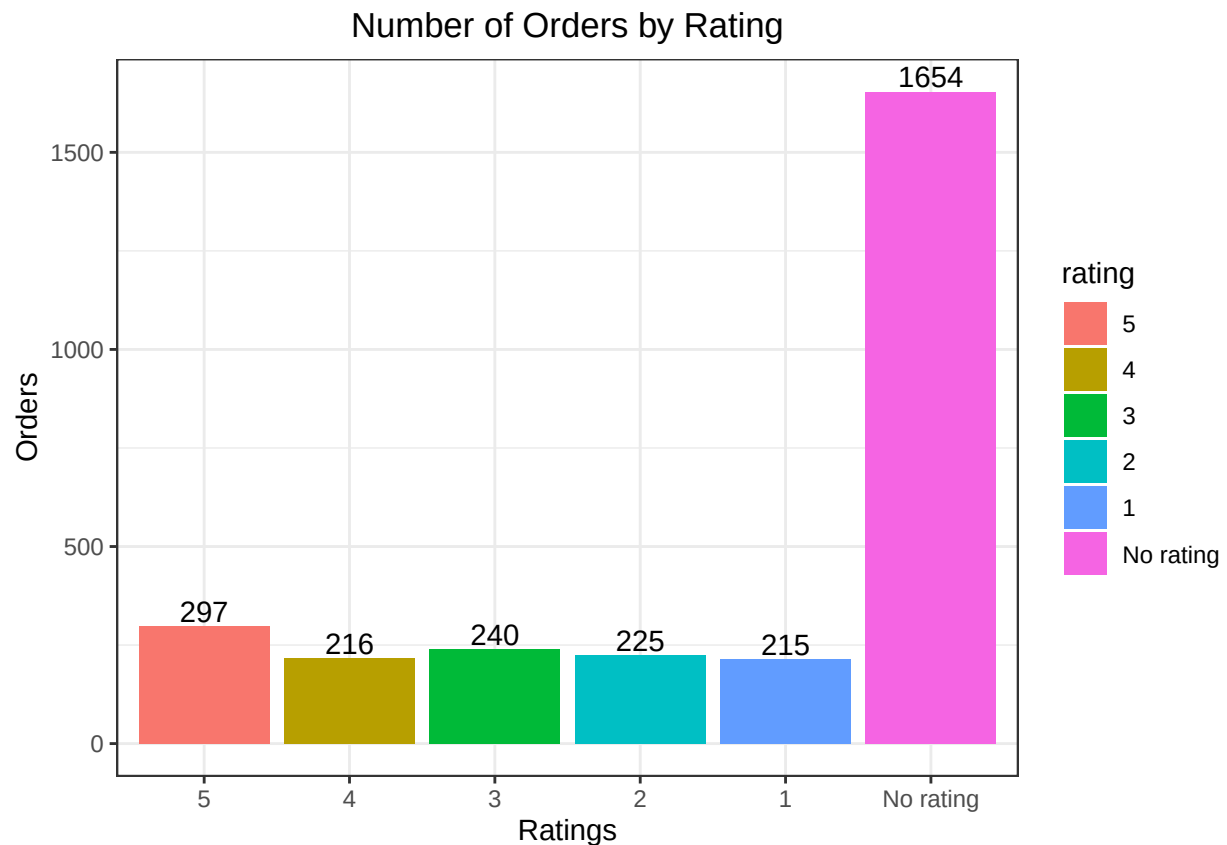
```

rating	orders_by_ratings	revenue_by_rating	profit_by_rating	profit_margin
No rating	1654	290285.23	183760.92	63.30
1	215	28596.75	16339.88	57.14
2	225	31838.67	19250.57	60.46

rating	orders_by_ratings	revenue_by_rating	profit_by_rating	profit_margin
3	240	38663.38	24209.14	62.62
4	216	29468.41	17303.72	58.72
5	297	40565.93	23957.63	59.06

7.1 Orders by rating

```
# Create a bar chart showing the number of orders by rating
ggplot(rating_distribution, aes(x = rating, y = orders_by_ratings, fill = rating)) +
  geom_bar(stat = "identity") + # Draw bars using the number of orders
  geom_text(aes(label = orders_by_ratings), # Add order count labels above bars
            vjust = -0.2) +
  theme_bw() + # Use a black-and-white theme
  labs(
    title = "Number of Orders by Rating", # Chart title
    x = "Ratings", # Label for x-axis
    y = "Orders" # Label for y-axis
  ) +
  theme(plot.title = element_text(hjust = 0.5)) # Center the chart title
```



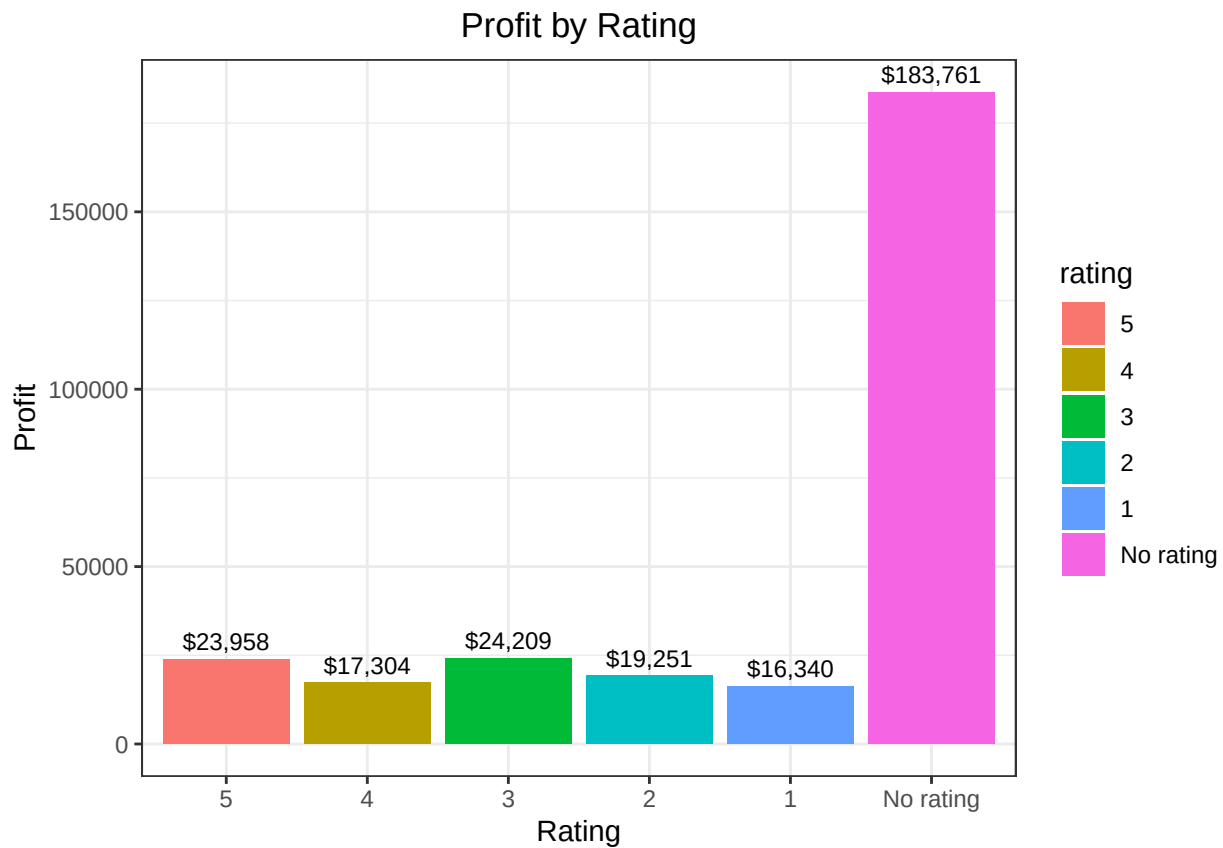
7.2 Profit by rating

```
# Create a bar chart showing profit by rating
ggplot(rating_distribution, aes(x = rating, y = profit_by_rating, fill = rating)) +
```

```

geom_bar(stat = "identity") + # Draw bars using profit values
geom_text(aes(label = dollar(profit_by_rating)), # Add profit labels above bars
          vjust = -0.5, size = 3) +
theme_bw() + # Use a black-and-white theme
labs(
  title = "Profit by Rating", # Chart title
  x = "Rating", # Label for x-axis
  y = "Profit" # Label for y-axis
) +
theme(plot.title = element_text(hjust = 0.5)) # Center the chart title

```

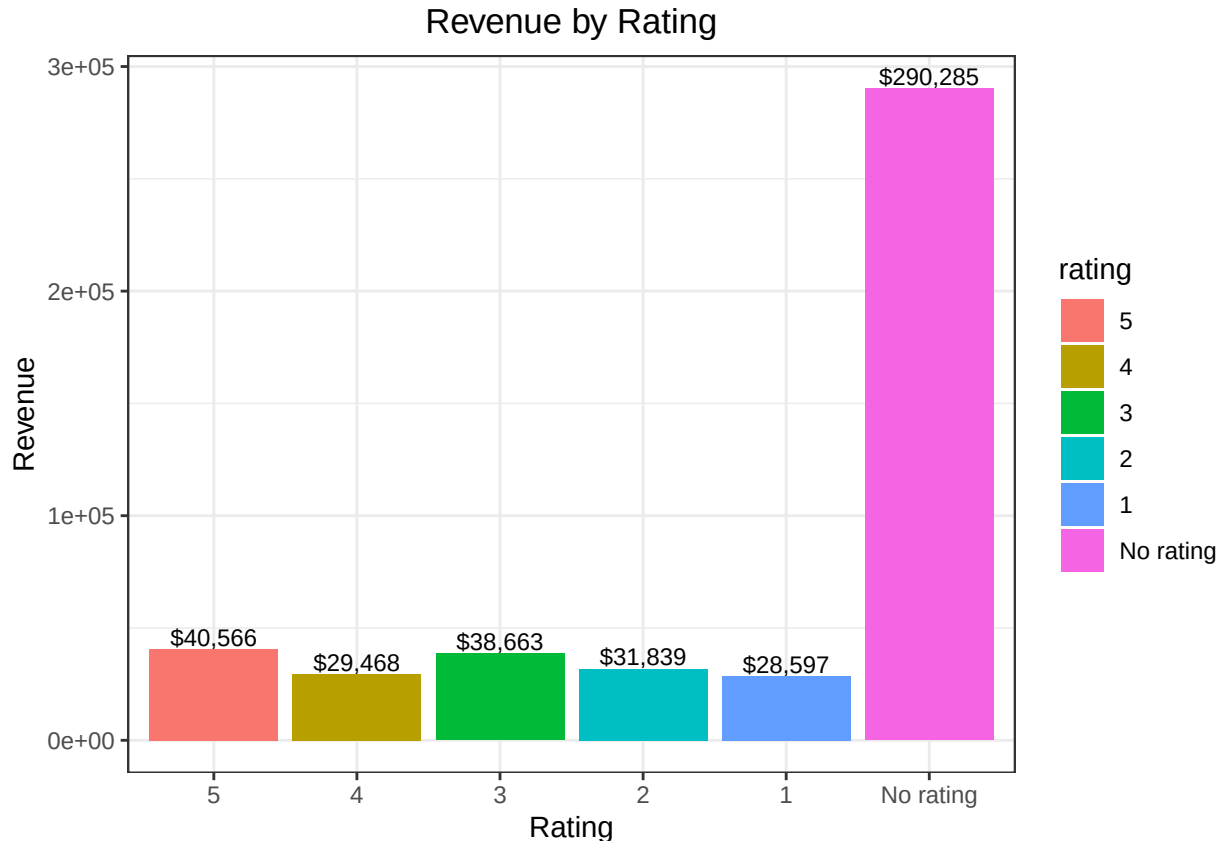


7.3 Revenue by rating

```

# Create a bar chart showing revenue by rating
ggplot(rating_distribution, aes(x = rating, y = revenue_by_rating, fill = rating)) +
  geom_bar(stat = "identity") + # Draw bars using revenue values
  geom_text(aes(label = dollar(revenue_by_rating)), # Add revenue labels above bars
            vjust = -0.2, size = 3.2) +
  theme_bw() + # Use a black-and-white theme
  labs(
    title = "Revenue by Rating", # Chart title
    x = "Rating", # Label for x-axis
    y = "Revenue" # Label for y-axis
  ) +
  theme(plot.title = element_text(hjust = 0.5)) # Center the chart title

```



7.4 Comments:

- **High number of unrated orders:**
The majority of orders (1,654) have no rating, indicating many customers do not provide feedback.
- **Unrated orders generate the highest revenue and profit:**
Despite lacking ratings, these orders account for the largest share of revenue (\$290K+) and profit (\$183K+).
- **Highest profit margin is from unrated orders (63.30%):**
This is followed by rating 3 orders at 62.62%, and rating 2 at 60.46%.
- **Profit margins do not increase linearly with rating:**
Rating 3 orders have higher profit margins than both ratings 4 and 5, which is unexpected.
- **7.5 Recommendation:**
Investigate product categories and pricing strategies by rating to better understand these patterns and improve customer feedback collection.

8) Analyze revenue, profit, and profit margin by state.

8.1 Best profitability efficiency (Top 3)

```
# Analyze profitability efficiency by state
# Join orders with customer data
profitability_efficiency <- inner_join(orders, customer, by = "customer_id") %>%
  group_by(State) %>% # Group the data by state
```

```

summarise(
  Revenue_by_state = sum(revenue),      # Total revenue for each state
  profit_by_state = sum(profit),        # Total profit for each state
  profit_margin = round((profit_by_state / Revenue_by_state) * 100, 2) # Profit margin (%)
) %>%
mutate(
  rank_by_margin = as.integer(dense_rank(desc(profit_margin))) # Rank states by profit margin
) %>%
arrange(rank_by_margin) %>% # Sort by rank (highest margin first)
filter(rank_by_margin <= 3) # Keep only the top 3 states
# Display the results as a table
kable(profiability_efficiency)

```

State	Revenue_by_state	profit_by_state	profit_margin	rank_by_margin
Utah	5256.80	3656.88	69.56	1
Massachusetts	8664.92	6023.10	69.51	2
New Mexico	2996.67	2044.45	68.22	3

8.2 Highest revenue and profit (top 3).

```

# Identify the top 3 states by profit and revenue
# Join orders with customer data
H_Revenue_Profit <- inner_join(orders, customer, by = "customer_id") %>%
  group_by(State) %>% # Group the data by state
  summarise(
    profit_state = sum(profit),      # Total profit for each state
    revenue_state = sum(revenue),    # Total revenue for each state
    profit_margin = round((profit_state / revenue_state) * 100, 2) # Profit margin (%)
  ) %>%
  arrange(desc(profit_state), desc(revenue_state)) %>% # Sort by highest profit and revenue
  mutate(rank = row_number()) %>% # Create ranking based on the sorted results
  filter(rank <= 3) # Keep only the top 3 states
# Show the results in a table
kable(H_Revenue_Profit)

```

State	profit_state	revenue_state	profit_margin	rank
California	34553.72	55470.47	62.29	1
Texas	32234.74	52305.67	61.63	2
Florida	22398.09	36250.92	61.79	3

8.3 least profitable and least revenue

```

# Identify the 3 states with the lowest profit and revenue
# Join orders with customer data
L_profitable_revenue <- inner_join(orders, customer, by = "customer_id") %>%
  group_by(State) %>% # Group the data by state
  summarise(
    profit_state = sum(profit),      # Total profit for each state
    revenue_state = sum(revenue),    # Total revenue for each state

```

```

    profit_margin = round((profit_state / revenue_state) * 100, 2) # Profit margin (%)
  ) %>%
  arrange(profit_state, revenue_state) %>% # Sort by lowest profit and revenue
  mutate(rank = row_number()) %>% # Assign rank based on sorted results
  filter(rank <= 3) # Keep the 3 states with the lowest profit
# Display the results as a table
kable(L_profitable_revenue)

```

State	profit_state	revenue_state	profit_margin	rank
Maine	16.72	91.10	18.35	1
Rhode Island	298.68	559.88	53.35	2
North Dakota	453.54	705.99	64.24	3

8.4 Smaller states like Delaware and New Hampshire show high margins despite lower total revenue.

```

# Find states ranked 4 and 5 by profit margin
# Join orders with customer data
small_States <- inner_join(orders, customer, by = "customer_id") %>%
  group_by(State) %>% # Group the data by state
  summarise(
    profit_state = sum(profit), # Total profit for each state
    revenue_state = sum(revenue), # Total revenue for each state
    profit_margin = round((profit_state / revenue_state) * 100, 2) # Profit margin (%)
  ) %>%
  arrange(desc(profit_margin)) %>% # Sort states by highest profit margin
  mutate(Rank = row_number()) %>% # Create ranking based on margin
  filter(Rank == 4 | Rank == 5) # Keep states ranked 4 and 5
# Show the results in a table
kable(small_States)

```

State	profit_state	revenue_state	profit_margin	Rank
Delaware	1658.72	2446.65	67.80	4
New Hampshire	1012.14	1497.36	67.59	5

8.5 Comments:

- Utah, Massachusetts, and New Mexico have the best profit efficiency.
- California, Texas, and Florida rank highest in both Revenue and Profit, but not in margin.
- Maine and Rhode Island are at the bottom in all three metrics: least profitable and least revenue.
- Smaller states like Delaware and New Hampshire show high margins despite lower total revenue.

8.6 Recommendation:

Focus on improving operational efficiency in large-volume states like California, Texas, and Florida to enhance profit margins. Additionally, investigate strategies to boost sales in underperforming states like Maine and Rhode Island, potentially through targeted promotions or revised product offerings.

9) Monthly profit trends and month-over-month comparisons.

I first create a new column called `Month_trend` by extracting the month number from the `New_date` column. Then, I replace those numeric month values with their full month names using `recode()`. This way, my `Month_trend` column is easier to understand because it shows names like “January” instead of just numbers.

```
orders<-orders %>%
  mutate(Month_trend= month(New_date))

Monthly_Trend<-orders %>%
  mutate(
    ## Convert numeric month codes to month names
    Month_trend=recode(Month_trend, "1"="January",
      "2"="February",
      "3"="March",
      "4"="April",
      "5"="May",
      "6"="June",
      "7"="July",
      "8"="August",
      "9"="September",
      "10"="October",
      "11"="November",
      "12"="December"),
    # Set factor levels to order months chronologically for plotting
    Month_trend=factor(Month_trend,
      levels = c("January",
        "February",
        "March",
        "April",
        "May",
        "June",
        "July",
        "August",
        "September",
        "October",
        "November",
        "December")) %>%
    group_by(Month_trend) %>% # Group data by Month_trend
    summarise(Monthly_Profit= sum(profit)) # Total profit per month

kable(Monthly_Trend) # View summarized monthly profit data
```

Month_trend	Monthly_Profit
January	14013.52
February	11244.50
March	18335.76
April	41131.00
May	38847.24

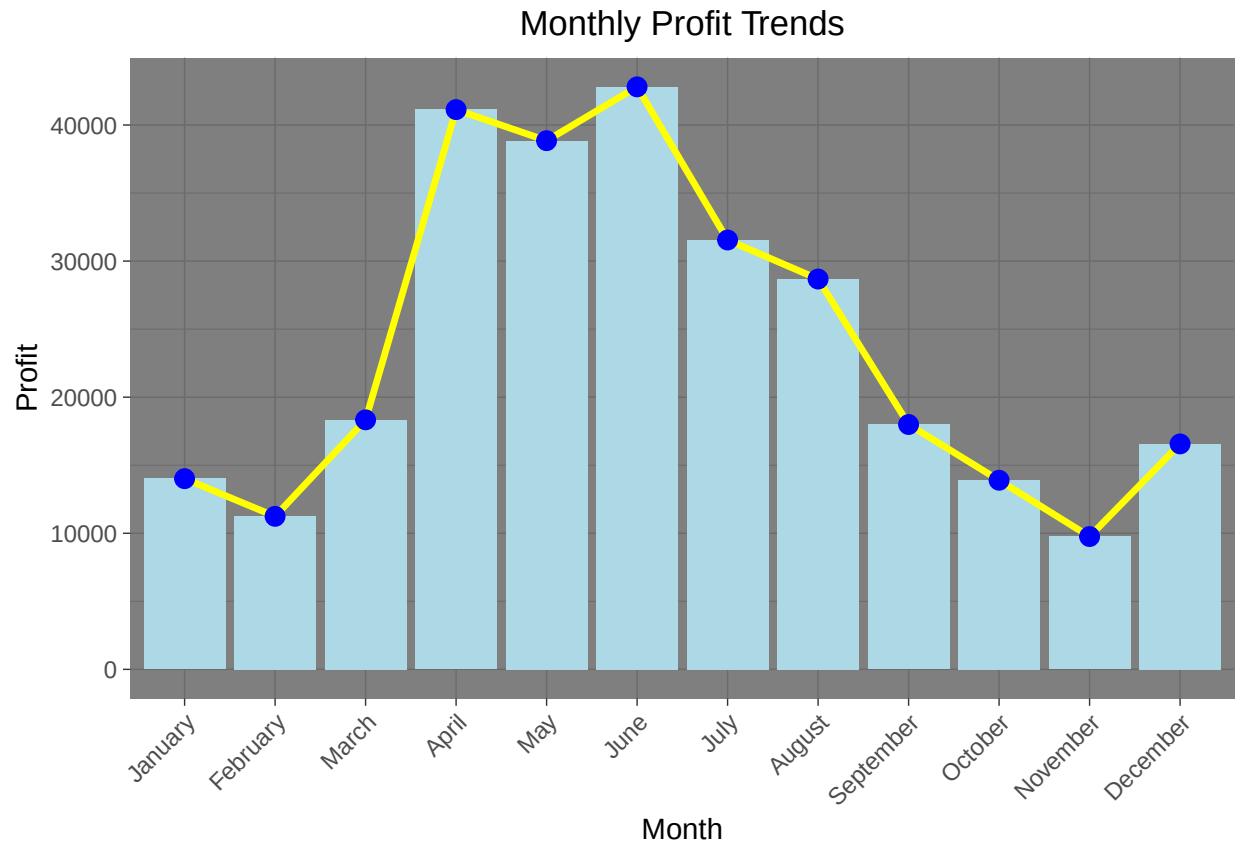
Month_trend	Monthly_Profit
June	42802.26
July	31550.40
August	28681.36
September	17992.00
October	13895.44
November	9760.52
December	16567.86

Monthly Profit Trend Graphic

```
# Create a chart showing monthly profit trends
ggplot(Monthly_Trend, aes(x = Month_trend, y = Monthly_Profit)) +
  geom_bar(stat = "identity", fill = "lightblue") + #Bars showing monthly profit
  geom_line(aes(group = 1), color = "yellow", size = 1.2) + #Line to show the trend
  geom_point(color = "blue", size = 3) + # Add points to highlight each month
  theme_dark() + # Use a dark theme for the chart
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1), # Rotate month labels
    plot.title = element_text(hjust = 0.5) # Center the chart title
  ) +

  labs(
    title = "Monthly Profit Trends", # Chart title
    x = "Month", # Label for x-axis
    y = "Profit" # Label for y-axis
  )
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



- June has the highest profit (\$42,802), marking a mid-year sales peak.
- April (\$41,131) and May (\$38,847) also show strong spring profits.
- January (\$14,014) and February (\$11,245) have low profits, likely due to post-holiday slowdowns.
- November has the lowest profit (\$9,761), possibly indicating inventory or sales issues despite the holiday season.
- Profit fluctuates month-to-month, with sharp increases from March to April and declines after July.
- July (\$31,550) dips compared to June, suggesting a mid-summer slowdown.
- September (\$17,992) and October (\$13,895) are relatively low, reflecting seasonal variation or challenges.
- December (\$16,568) rebounds after November but stays below spring/summer levels.
- Overall, there is clear seasonality with peaks in late spring/early summer and dips in late fall/early year.

10) Monthly Revenue trends and month-over-month comparisons.

```
Revenue_Trend<-orders %>%
  mutate(
    ## Convert numeric month codes to month names
```

```

Month_trend =recode( Month_trend,
  "1"="January",
  "2"="February",
  "3"="March",
  "4"="April",
  "5"="May",
  "6"="June",
  "7"="July",
  "8"="August",
  "9"="September",
  "10"="October",
  "11"="November",
  "12"="December"),
# Set factor levels to order months chronologically for plotting
  Month_trend=factor(Month_trend, levels= c("January",
  "February",
  "March",
  "April",
  "May",
  "June",
  "July",
  "August",
  "September",
  "October",
  "November",
  "December")) %>%
  group_by(Month_trend) %>% # Group data by Month_trend
  summarise(Revenue= sum(revenue)) # Calculate total revenue per month

Revenue_Trend

```

```

## # A tibble: 12 x 2
##   Month_trend Revenue
##   <fct>         <dbl>
## 1 January      35283.
## 2 February    27718.
## 3 March       31311.
## 4 April       55438.
## 5 May         55082.
## 6 June        56407.
## 7 July        50390.
## 8 August      45469.
## 9 September   33367.
## 10 October    27995.
## 11 November   17088.
## 12 December   23870.

```

Monthly Revenue Trend Graphic

```

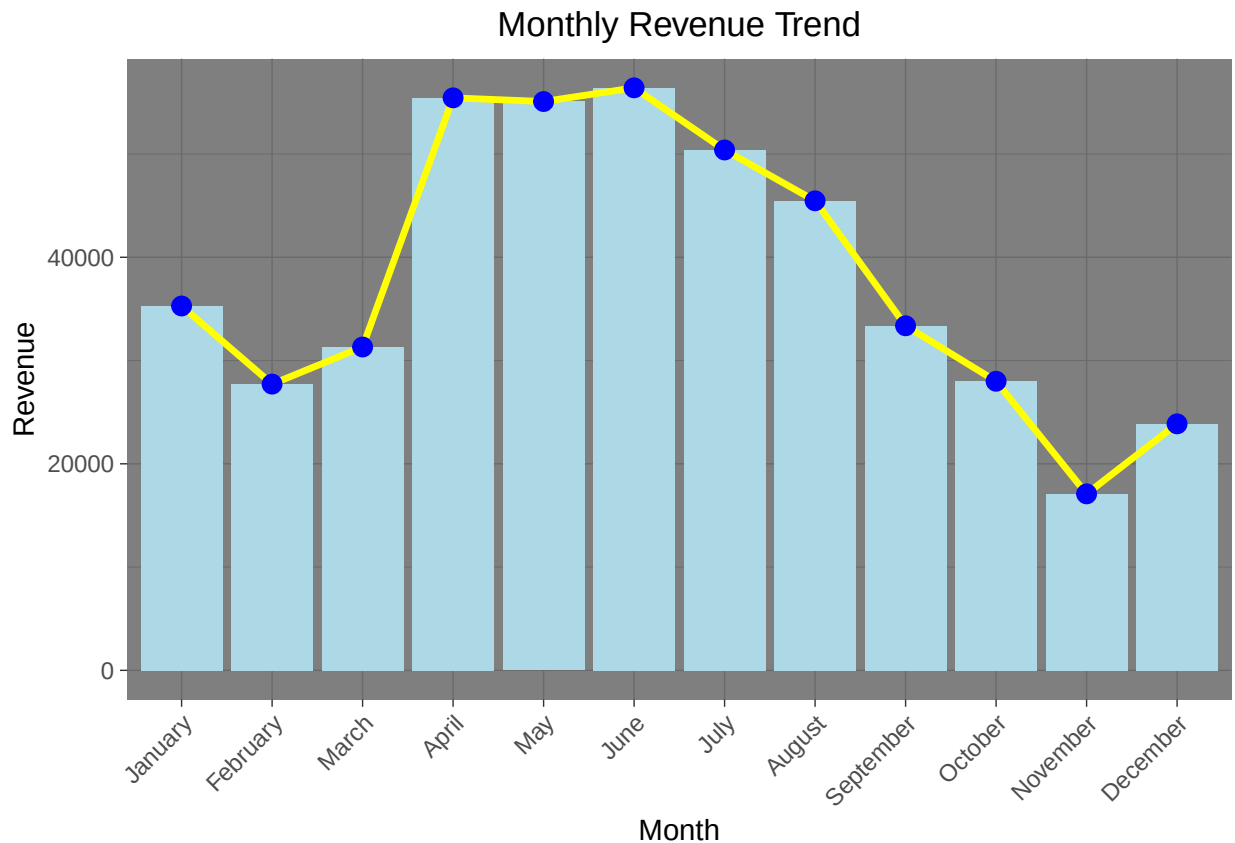
# Create a chart showing the monthly revenue trend
ggplot(Revenue_Trend, aes(x = Month_trend, y = Revenue)) +
  geom_bar(stat = "identity", fill = "lightblue") + # Create bars showing revenue per month
  geom_line(aes(group = 1), color = "yellow", size = 1.2) + # Line to show the trend
  geom_point(color = "blue", size = 3) + # Add points for each month

```

```

theme_dark() + # Use a dark theme for the chart
theme(
  axis.text.x = element_text(angle = 45, hjust = 1), # Rotate month labels
  plot.title = element_text(hjust = 0.5) # Center the chart title
) +
labs(
  title = "Monthly Revenue Trend", # Chart title
  x = "Month", # Label for x-axis
  y = "Revenue" # Label for y-axis
)

```



- June had the highest revenue (\$56,407), marking peak business in early summer.

- April (\$55,438) and May (\$55,082) showed strong spring revenue.
- July (\$50,390) and August (\$45,469) maintained solid summer performance.
- November recorded the lowest revenue (\$17,088), unusually low despite events like Black Friday.
- December revenue (\$23,870) improved from November but stayed below yearly average.
- January to March had modest revenues (\$27K–\$35K), indicating a slow start.
- Sharp increase from March to April suggests seasonal growth or successful campaigns.
- Revenue declined gradually from September (\$33,367) to October (\$27,995), showing post-summer slowdown.
- Overall, the data reveals clear seasonality with strong spring and early summer performance.

11) Executive Summary

The business demonstrates strong profitability with an average profit margin of 62%, highlighting sports such as soccer and basketball for their volume and profit margins.

Customer ratings do not show a direct correlation with profitability; however, low ratings indicate potential areas for improvement.

Geographically, states like Utah and Massachusetts stand out for margin efficiency, while high-volume states such as California, Texas, and Florida dominate in total revenue and profit.

Seasonality is evident, with peaks in spring and summer (April to June) and notable declines in winter and fall, especially in November.

It is recommended to focus marketing campaigns during peak months and investigate the causes of low profitability in underperforming states and months.